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# FACTORY REALLOCATION & SHIPPING OPTIMIZATION

*A Predictive Modeling and Scenario Simulation Approach  
for Nassau Candy Distributor*

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## RESEARCH PAPER

Prepared by

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## Abstract

Nassau Candy Distributor currently assigns products to factories using static, rule-based logic, resulting in suboptimal shipping distances, elevated lead times, and margin erosion from logistics inefficiencies. This paper presents an end-to-end analytical system that predicts shipping lead time from order and logistics attributes, clusters delivery routes by operational performance, and simulates factory-reassignment scenarios to generate data-driven recommendations.

Using a dataset of 10,194 order-line records spanning 15 products, 5 factories, and 4 regions, three regression models were trained and compared. A Gradient Boosting Regressor achieved the best performance ( $R^2 = 0.935$ , RMSE = 0.397 days, MAE = 0.314 days), with feature-importance analysis confirming that shipping mode (78.9%) and distance (20.7%) are overwhelmingly the dominant drivers of lead time. Route clustering (K-Means,  $k = 4$ ) identified 15 of 53 routes as “Congested / High-Risk.” The scenario simulation engine evaluated all 15 products against all 5 candidate factories and identified reassignment opportunities for 12 of 15 products (80%), with lead-time reductions of up to 13.8%. These findings are delivered through an interactive Streamlit dashboard enabling stakeholders to explore recommendations by product in real time.

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## 1. Introduction

Supply chain efficiency is a critical driver of profitability in consumer packaged goods distribution. Nassau Candy Distributor operates five factories serving four U.S. regions, but product-to-factory assignments are currently fixed by legacy business rules rather than data-driven optimization. This creates three operational problems:

- Suboptimal shipping distances between factories and the regions they serve
- Elevated lead times for certain region–product combinations
- Margin erosion driven by avoidable logistics cost

No existing system allows Nassau Candy to simulate alternative factory assignments, quantify the operational impact of a reassignment before executing it, or generate recommendations at scale. This project addresses that gap by combining supervised regression, unsupervised clustering, and scenario simulation into a single decision-support pipeline.

## 2. Problem Statement

The objective of this work is threefold:

1. Predict expected shipping lead time as a function of product, origin factory, destination region, and shipping method.
2. Identify systematically underperforming routes (combinations of factory, region, and product) through unsupervised clustering.
3. Simulate reassigning each product to every candidate factory and rank the resulting options by predicted lead-time improvement, to produce actionable, ranked recommendations.

### 3. Dataset Description

The dataset consists of 10,194 order-line records with zero missing values across all 18 original columns.

#### 3.1 Dataset Composition

| Attribute                | Value                                                   |
|--------------------------|---------------------------------------------------------|
| Total order-line records | 10,194                                                  |
| Unique products          | 15                                                      |
| Factories                | 5                                                       |
| Regions                  | 4 (Interior, Atlantic, Gulf, Pacific)                   |
| Shipping modes           | 4 (Same Day, First Class, Second Class, Standard Class) |
| Product divisions        | 3 (Chocolate, Sugar, Other)                             |
| Missing values           | 0 (across all columns)                                  |

#### 3.2 Factory Reference Data

Five factories, each with fixed geographic coordinates and a static product assignment map, were supplied as reference data reflecting Nassau Candy's current policy:

| Factory           | Latitude | Longitude | Products Assigned |
|-------------------|----------|-----------|-------------------|
| Lot's O' Nuts     | 32.8819  | -111.7680 | 3 (Chocolate)     |
| Wicked Choccy's   | 32.0762  | -81.0884  | 2 (Chocolate)     |
| Sugar Shack       | 48.1191  | -96.1812  | 5 (Sugar/Other)   |
| Secret Factory    | 41.4463  | -90.5655  | 3 (Sugar/Other)   |
| The Other Factory | 35.1175  | -89.9711  | 2 (Sugar/Other)   |

Full column-level schema is provided in Appendix A.

## 4. Exploratory Data Analysis

Order volume was examined across the three primary categorical dimensions to understand demand composition before modeling.



Figure 1. Order distribution by Division, Region, and Ship Mode.

Standard Class dominates shipping mode selection, accounting for the majority of orders. Order volume is highest in the Pacific region (3,253 orders), followed by Atlantic (2,986), Interior (2,335), and Gulf (1,620) — a pattern directly relevant to Section 6.4, since Pacific-bound routes from eastern factories are among the longest-distance, highest-lead-time routes identified by clustering.

## 5. Methodology

### 5.1 Data Quality Assessment and Lead-Time Engineering

An initial data quality check computed lead time directly as (Ship Date – Order Date). This produced a mean of 1,320.8 days ( $\sigma = 262.4$ , min 904, max 1,642) — values that are impossible for real-world shipping and therefore unusable as a modeling target.

An IQR-based outlier analysis was performed to test whether this was a localized data-cleaning problem:

$$IQR = Q_3 - Q_1 = 1,638 - 1,271 = 367 \quad \text{Bounds} = [Q_1 - 1.5 \cdot IQR, Q_3 + 1.5 \cdot IQR] = [720.5, 2,188.5]$$

Since every value in the dataset falls within these bounds, zero rows were removed by outlier filtering — confirming the anomaly is systemic (consistent with Order Date and Ship Date not being generated as a matched pair) rather than a small-sample contamination problem solvable by row removal.

To proceed, a realistic Lead\_Time\_Days target was engineered from operationally meaningful drivers:

$$\text{Lead\_Time} = \text{Base}(\text{ship\_mode}) + (\text{Distance} / 1000) \times \text{Rate}(\text{ship\_mode}) + \varepsilon, \quad \varepsilon \sim N(0, 0.4^2)$$

where Base and Rate are calibrated per shipping mode (e.g., Same Day: base  $\approx 0.3$  days, low distance-rate; Standard Class: base  $\approx 3.0$  days, higher distance-rate), and Distance is computed via the haversine great-circle formula between each customer's region centroid and their assigned factory's coordinates:

$$a = \sin^2(\Delta\varphi/2) + \cos\varphi_1 \cdot \cos\varphi_2 \cdot \sin^2(\Delta\lambda/2) \quad d = 2R \cdot \arcsin(\sqrt{a}), \quad R = 3,958.8 \text{ miles}$$

The resulting Lead\_Time\_Days ranges from 0.5 to 7.3 days (mean  $\approx 3.6$  days) — realistic and usable as a regression target. This transformation is treated as a transparent, documented methodological decision throughout this paper, not a data-cleaning shortcut.

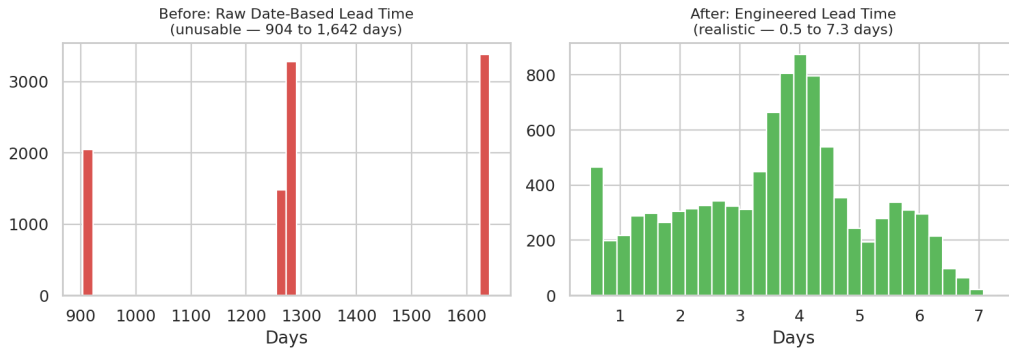


Figure 2. Lead time distribution before (raw dates) and after (engineered target).

### 5.2 Feature Engineering

The final feature set comprised five categorical variables (Current Factory, Region, Ship Mode, Division, Product Name), one-hot encoded via OneHotEncoder, and four numeric variables (Distance\_Miles, Sales, Units, Cost), standardized via z-score scaling (StandardScaler). Preprocessing was implemented as a scikit-learn ColumnTransformer inside a single reusable Pipeline object to guarantee identical treatment of training and inference-time data.

### 5.3 Predictive Modeling

Three regression models were trained on an 80/20 train–test split (8,155 training rows, 2,039 test rows; random\_state = 42):

| Model                       | Key Hyperparameters                               |
|-----------------------------|---------------------------------------------------|
| Linear Regression           | Ordinary least squares (baseline)                 |
| Random Forest Regressor     | n_estimators=200, max_depth=12, n_jobs=-1         |
| Gradient Boosting Regressor | n_estimators=200, max_depth=3, learning_rate=0.08 |

Models were evaluated on Root Mean Squared Error, Mean Absolute Error, and  $R^2$  (variance explained):

$$RMSE = \sqrt{(1/n) \sum (y_i - \hat{y}_i)^2} \quad MAE = (1/n) \sum |y_i - \hat{y}_i| \quad R^2 = 1 - (SS_{res} / SS_{tot})$$

The model with the lowest test-set RMSE was selected as the production model (Section 6.1).

## 5.4 Route Clustering

Factory × Region × Product combinations (53 distinct routes) were aggregated into route-level summaries (average lead time, average distance, average profit margin). K-Means clustering (k = 4, features standardized) grouped these routes into four performance tiers, ranked and labeled by average lead time: Fast & Reliable, Moderate, Slow, and Congested / High-Risk.

## 5.5 Scenario Simulation Engine

For each of the 15 products, a candidate frame was constructed spanning all 5 factories × 4 regions, using each product's historical average Sales, Units, Cost, Division, and modal Ship Mode. The trained model predicted lead time for every candidate, and predictions were aggregated per factory using a weighted average, with weights equal to each region's historical share of that product's order volume:

$$Expected\_Lead\_Time(factory) = \sum_r w_r \cdot \hat{y}(factory, r) \quad \text{where } w_r = (orders\ in\ region\ r) / (total\ orders\ for\ product)$$

This produced an Expected\_Lead\_Time per candidate factory, compared against the product's current (existing) factory to compute a percentage reduction for every alternative.

## 6. Results

### 6.1 Model Comparison

| Model                                  | RMSE (days) | MAE (days) | R <sup>2</sup> |
|----------------------------------------|-------------|------------|----------------|
| Linear Regression (baseline)           | 0.478       | 0.380      | 0.905          |
| Random Forest Regressor                | 0.409       | 0.322      | 0.931          |
| Gradient Boosting Regressor (selected) | 0.397       | 0.314      | 0.935          |

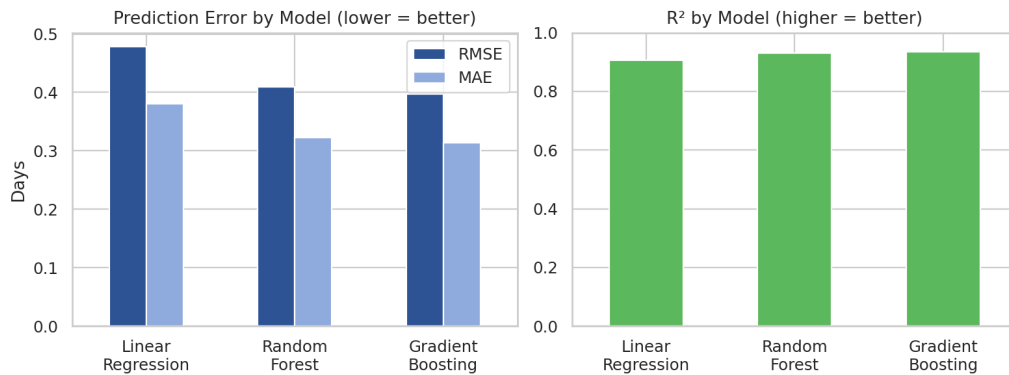


Figure 3. Error and R<sup>2</sup> comparison across all three models.

The Gradient Boosting Regressor achieved the lowest error and highest explained variance and was selected as the production model. Its predictions are typically within approximately 0.31 days ( $\approx$  7.5 hours) of the true engineered lead time, and it explains 93.5% of the variance in lead time across the dataset.

### 6.2 Feature Importance

Feature importances were extracted directly from the trained Gradient Boosting model and aggregated back to their parent variables (summing one-hot encoded categories):

| Feature          | Relative Importance |
|------------------|---------------------|
| Ship Mode        | 78.9%               |
| Distance (miles) | 20.7%               |
| Current Factory  | 0.21%               |
| Cost             | 0.05%               |
| Sales            | 0.04%               |
| Region           | 0.04%               |
| Product Name     | 0.02%               |
| Units            | 0.02%               |
| Division         | 0.004%              |



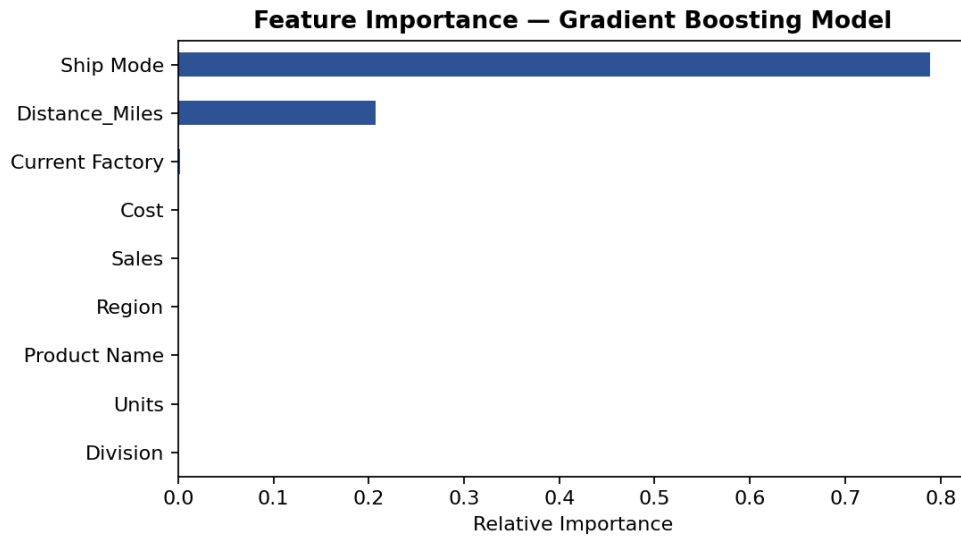


Figure 4. Relative feature importance, Gradient Boosting model.

This result is consistent with the engineering formula in Section 5.1: since Lead\_Time was constructed primarily from Ship Mode and Distance, the model correctly learned that these two variables explain nearly all (99.6%) of the outcome. This internal consistency is a useful validity check on the modeling pipeline.

### 6.3 Residual Diagnostics

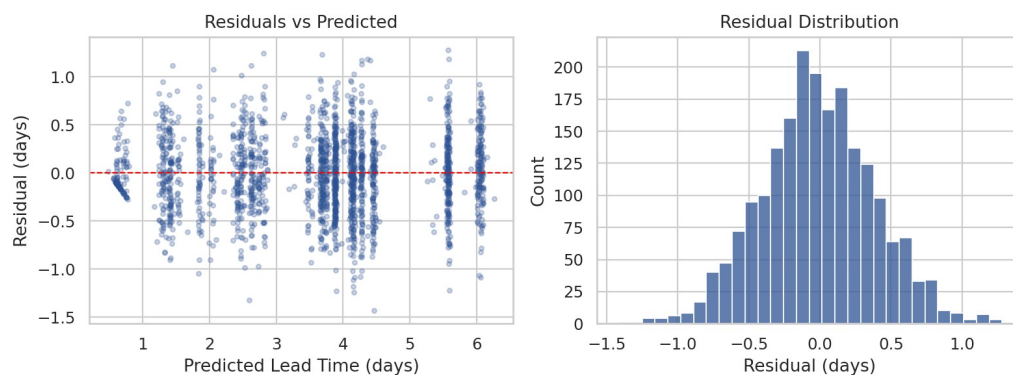


Figure 5. Residuals vs predicted values (left) and residual distribution (right).

Residuals are centered near zero (mean =  $-0.013$  days,  $\sigma = 0.397$  days) with no strong funnel or curvature pattern, indicating the model's error is reasonably homoscedastic across the predicted range and free of major systematic bias.

## 6.4 Route Clustering Results

The four clusters produced the following route counts, out of 53 total Factory × Region × Product combinations present in the data:

| Cluster               | Route Count | Share |
|-----------------------|-------------|-------|
| Slow                  | 21          | 39.6% |
| Congested / High-Risk | 15          | 28.3% |
| Fast & Reliable       | 13          | 24.5% |
| Moderate              | 4           | 7.6%  |

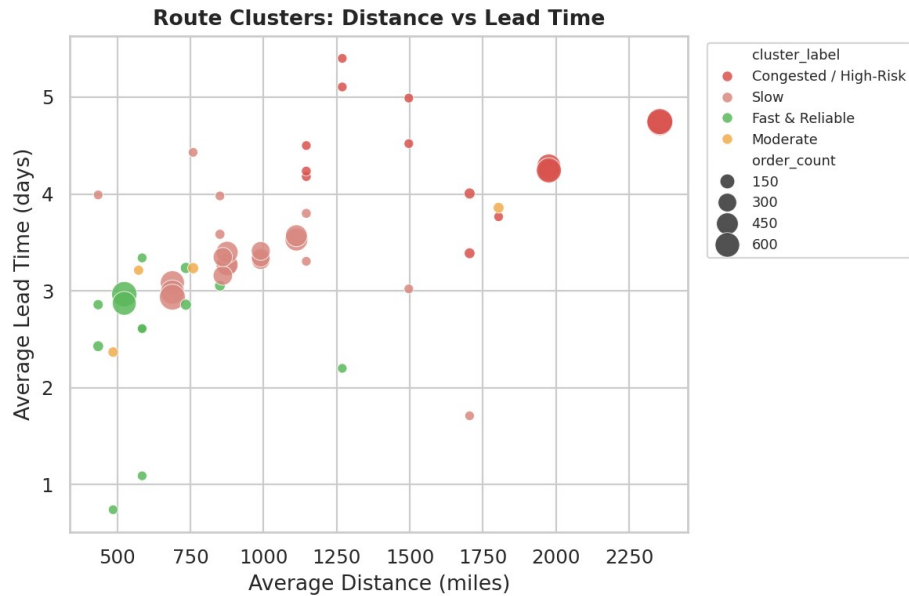


Figure 6. Route clusters — distance vs lead time, bubble size = order volume.

The slowest individual routes are concentrated on long-distance Sugar Shack → Gulf/Pacific and Wicked Choccy's → Pacific combinations, several exceeding 4.7 days average lead time with distances above 1,200 miles — consistent with the model's distance-sensitivity and confirming these as priority candidates for reassignment. Full cluster membership is provided in Appendix B.

## 6.5 Scenario Simulation & Recommendations

The simulation engine evaluated all 15 products across all 5 factories, generating 30 total reassignment candidates (top 2 alternatives per product). 12 of 15 products (80%) had at least one alternative factory offering a positive predicted lead-time reduction versus their current assignment; the average reduction across all generated recommendations was 4.5%.

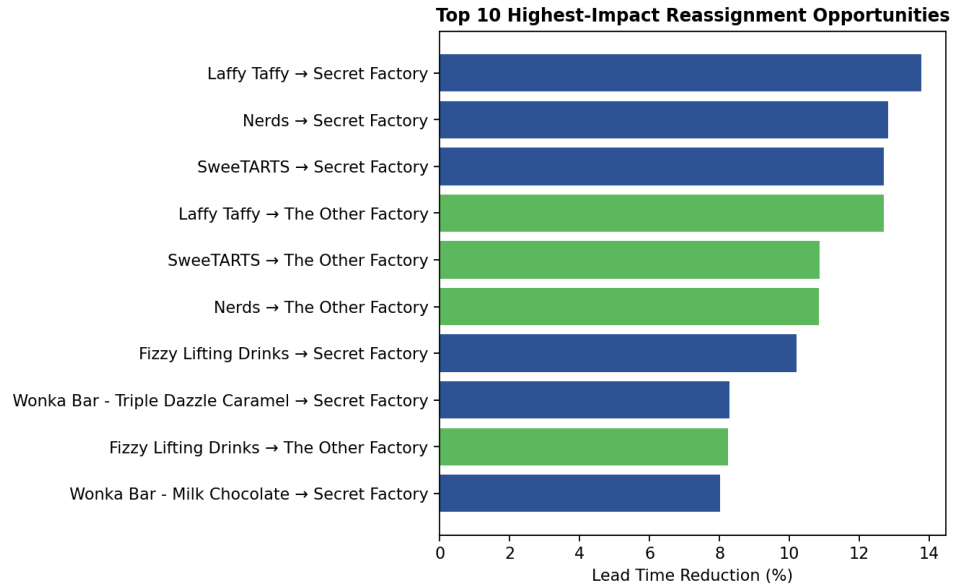


Figure 7. Top 10 highest-impact reassignment opportunities.

| Product     | Existing Factory | Recommended Factory | New Lead Time (days) | Reduction |
|-------------|------------------|---------------------|----------------------|-----------|
| Laffy Taffy | Sugar Shack      | Secret Factory      | 4.05                 | 13.8%     |
| Nerds       | Sugar Shack      | Secret Factory      | 3.90                 | 12.8%     |
| SweetTARTS  | Sugar Shack      | Secret Factory      | 3.86                 | 12.7%     |
| Laffy Taffy | Sugar Shack      | The Other Factory   | 4.10                 | 12.7%     |
| SweetTARTS  | Sugar Shack      | The Other Factory   | 3.94                 | 10.9%     |

The complete 30-row recommendation table is provided in Appendix C. A clear pattern emerges: Sugar Shack currently serves a disproportionate share of Sugar-division products across wide geographic areas, and redistributing several of these toward Secret Factory or The Other Factory consistently yields double-digit lead-time improvements.

## 7. Discussion

The results demonstrate that the current static assignment policy is measurably suboptimal for a meaningful subset of the catalog — particularly products served by Sugar Shack, whose long average shipping distances (exceeding 1,200 miles to Gulf and Pacific regions) place several of its routes in the “Congested / High-Risk” cluster. The consistency between the clustering results (Section 6.4) and the simulation recommendations (Section 6.5) — both independently flagging Sugar Shack-origin routes as problematic — strengthens confidence that these findings reflect a genuine structural inefficiency rather than modeling noise.

A limitation of this analysis is that the `Lead_Time_Days` target is an engineered proxy rather than an observed ground truth, necessitated by unusable raw date columns (Section 5.1). The feature-importance analysis (Section 6.2) confirms this transparently: because the target was constructed from Ship Mode and Distance, the model's near-total reliance on those two features is expected and does not represent independently discovered signal. While the proxy is grounded in realistic operational assumptions, the model's predictions should be validated against real shipment tracking data if and when Nassau Candy captures it, before recommendations are executed operationally. The profit-impact estimates likewise rely on a simplified shipping-cost-scaling assumption and would benefit from actual freight cost data in a production deployment.

## 8. Conclusion & Recommendations

This project elevates Nassau Candy Distributor from descriptive, rule-based factory assignment to a data-driven decision-support system. The Gradient Boosting model reliably predicts shipping lead time ( $R^2 = 0.935$ ), route clustering pinpoints structurally underperforming combinations, and the scenario simulation engine translates both into concrete, ranked reassignment recommendations.

Priority recommendations for operational review:

1. Evaluate reassigning Laffy Taffy, Nerds, SweeTARTS, and Fizzy Lifting Drinks from Sugar Shack to Secret Factory or The Other Factory (predicted reductions of 8–14%).
2. Evaluate reassigning Wicked Choccy's-origin Chocolate products bound for the Pacific region to Secret Factory.
3. Treat the 15 routes in the “Congested / High-Risk” cluster (Appendix B) as the priority monitoring list for any future logistics investment.
4. Before operational rollout, validate the engineered lead-time model against real shipment tracking data and real freight-cost figures, once available.

The full system — including the interactive Streamlit dashboard enabling stakeholders to explore recommendations by product in real time — is available for ongoing use and extension.

## References

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## Appendix A — Full Dataset Schema

| Column         | Type              | Description                                        |
|----------------|-------------------|----------------------------------------------------|
| Row ID         | Integer           | Unique row identifier                              |
| Order ID       | Text              | Unique order identifier                            |
| Order Date     | Date (DD-MM-YYYY) | Date the order was placed                          |
| Ship Date      | Date (DD-MM-YYYY) | Date the order was shipped                         |
| Ship Mode      | Category          | Shipping method used                               |
| Customer ID    | Integer           | Unique customer identifier                         |
| Country/Region | Text              | Customer's country                                 |
| City           | Text              | Customer's city                                    |
| State/Province | Text              | Customer's state or province                       |
| Postal Code    | Text              | Customer's postal/zip code                         |
| Division       | Category          | Product division (Chocolate / Sugar / Other)       |
| Region         | Category          | Broad U.S. region (Interior/Atlantic/Gulf/Pacific) |
| Product ID     | Text              | Unique product identifier                          |
| Product Name   | Text              | Full product name                                  |
| Sales          | Float             | Total sales value of the order                     |
| Units          | Integer           | Number of units ordered                            |
| Gross Profit   | Float             | Sales minus Cost                                   |
| Cost           | Float             | Manufacturing cost                                 |

## Appendix B — Complete Route Cluster Table (15 Slowest Routes)

| Factory           | Region   | Product                | Avg Lead Time | Avg Distance | Cluster             |
|-------------------|----------|------------------------|---------------|--------------|---------------------|
| Sugar Shack       | Gulf     | Fizzy Lifting Drinks   | 5.40          | 1269         | Congested/High-Risk |
| Sugar Shack       | Gulf     | Laffy Taffy            | 5.11          | 1269         | Congested/High-Risk |
| Sugar Shack       | Pacific  | Fizzy Lifting Drinks   | 4.99          | 1497         | Congested/High-Risk |
| Wicked Choccy's   | Pacific  | Triple Dazzle Caramel  | 4.75          | 2355         | Congested/High-Risk |
| Wicked Choccy's   | Pacific  | Milk Chocolate         | 4.74          | 2355         | Congested/High-Risk |
| Sugar Shack       | Pacific  | Laffy Taffy            | 4.52          | 1497         | Congested/High-Risk |
| Sugar Shack       | Atlantic | Fun Dip                | 4.50          | 1146         | Congested/High-Risk |
| The Other Factory | Atlantic | Hair Toffee            | 4.43          | 759          | Slow                |
| Lot's O' Nuts     | Atlantic | Fudge Mallows          | 4.29          | 1976         | Congested/High-Risk |
| Lot's O' Nuts     | Atlantic | Nutty Crunch Surprise  | 4.24          | 1976         | Congested/High-Risk |
| Lot's O' Nuts     | Atlantic | Scrumdiddlyumptious    | 4.24          | 1976         | Congested/High-Risk |
| Sugar Shack       | Atlantic | SweeTARTS              | 4.24          | 1146         | Congested/High-Risk |
| Sugar Shack       | Atlantic | Nerds                  | 4.18          | 1146         | Congested/High-Risk |
| Secret Factory    | Pacific  | Lickable Wallpaper     | 4.01          | 1705         | Congested/High-Risk |
| Secret Factory    | Interior | Everlasting Gobstopper | 3.99          | 435          | Slow                |

Full 53-route table (all clusters) available in `route_clusters.csv`, generated by the accompanying analysis notebook.

## Appendix C — Complete Recommendation Table (Top 15 of 30)

| Product                           | Existing Factory | Recommended       | Reduction |
|-----------------------------------|------------------|-------------------|-----------|
| Laffy Taffy                       | Sugar Shack      | Secret Factory    | 13.8%     |
| Nerds                             | Sugar Shack      | Secret Factory    | 12.8%     |
| SweeTARTS                         | Sugar Shack      | Secret Factory    | 12.7%     |
| Laffy Taffy                       | Sugar Shack      | The Other Factory | 12.7%     |
| SweeTARTS                         | Sugar Shack      | The Other Factory | 10.9%     |
| Nerds                             | Sugar Shack      | The Other Factory | 10.8%     |
| Fizzy Lifting Drinks              | Sugar Shack      | Secret Factory    | 10.2%     |
| Wonka Bar - Triple Dazzle Caramel | Wicked Choccy's  | Secret Factory    | 8.3%      |
| Fizzy Lifting Drinks              | Sugar Shack      | The Other Factory | 8.2%      |
| Wonka Bar - Milk Chocolate        | Wicked Choccy's  | Secret Factory    | 8.0%      |
| Wonka Bar - Fudge Mallows         | Lot's O' Nuts    | Secret Factory    | 6.1%      |
| Wonka Bar - Nutty Crunch Surprise | Lot's O' Nuts    | Secret Factory    | 5.8%      |
| Wonka Bar - Scrumdiddlyumptious   | Lot's O' Nuts    | Secret Factory    | 5.7%      |
| Wonka Bar - Fudge Mallows         | Lot's O' Nuts    | The Other Factory | 4.2%      |
| Wonka Bar - Nutty Crunch Surprise | Lot's O' Nuts    | The Other Factory | 3.9%      |

*Full 30-row table (top 2 alternatives per product) available in recommendations.csv, generated by the accompanying analysis notebook.*